Cascading Stability Review Report

order-api

Case ID	BZ-20260422-4423
Severity	sev2
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BUFFERZONE
AETERNA ENGINE · POST-INCIDENT REVIEW

Executive Summary

The materials in this case do not include a distinct business-impact narrative.

QUANTITATIVE SIGNAL EXCERPTS (FROM THE CASE RECORD)

- L: [Signals] metrics, alerts - p99_latency_ms: 1380 (threshold 500) ms threshold source: 输入提供 - error_ratio: 0.0 (threshold 0.01) ratio threshold source: 输入提供 [Over threshold] - p99_latency_ms: 1380 > 500 - error_ratio: ...
- - L-RT-001: p99_latency_ms obs=1380 thr=500 matched=True (metrics) - L-RT-002: error_ratio obs=0.0 thr=0.01 matched=True (metrics) - L-RT-ALR-001: (alerts) obs=1 thr=0 matched=True (alerts)
- Q: [Signals] metrics, alerts - queue_depth: 1204 (threshold 800) count threshold source: 输入提供 [Over threshold] - queue_depth: 1204 > 800 [] variable=Q level=warning score=40.1 rules=Q-RT-001 confidence: 队列/积压类指标与有效阈值比对
- 根因提示(语义):['traffic surge during promotion window', 'connection pool saturation', 'retry amplification under downstream timeout', 'shared resource contention'] [Signals] metrics, alerts, inferred_path, root_cause_hint - r...

BOTTOM-LINE ASSESSMENT

For this review, the incident is classified at sev2 severity. The structural assessment places elevated risk on L, B, P; these dimensions warrant priority attention. The documented cascade path and primary variable signals support a coherent management-level reading of propagation. On remediation sequencing, inner-loop work should lead with: Add circuit breaking, retry backoff, and concurrency limits at the stressed dependency boundary.

THREE ACTION PRIORITIES

1. **Prioritize the inner loop**
   implement Add circuit breaking, retry backoff, and concurrency limits at the stressed dependency boundary first to interrupt the positive-feedback propagation path.

2. **Then strengthen the outer loop**
   advance Design isolation around contention-prone workloads and shared-resource pressure to reduce repeated reliance on shared resources as propagation hubs.

3. **Resource focus**
   prioritize investment in buffering, sacrifice, and isolation points rather than relying solely on post-incident capacity scaling.

EXPECTED OUTCOME

The materials in this case do not include a distinct recovery narrative.

For management review — not a validated runbook.

Incident overview

Business Impact

The materials in this case do not include a distinct business-impact narrative.

User Impact

The materials in this case do not include a distinct user-impact narrative.

Five-variable structure

L CRITICAL
Core Continuity
Composite score：69
核心连续性(L)判为严重:延迟/错误类共 2 项超过有效阈值,关键项包括 p99_latency_ms、error_ratio 等。.

B CRITICAL
Buffer Capacity
Composite score：75
Current evidence shows materially elevated pressure on buffer capacity, and the B display is kept at an elevated-risk posture under evidence boundaries.

Q WARNING
Queue and Accumulation Pressure
Composite score：40.1
压力堆积(Q)判为警告:队列/积压类 1 项超阈值,以queue_depth 等为代表,堆积压力升高。.

X WARNING
Selective Sacrifice Intensity
Composite score：48
X (Localized yield intensity) is rated "Warning". The overall rating remains “Warning”.

P CRITICAL
Cascade Propagation Risk
Composite score：85
Focus service context: “order-api”. The cascade-propagation assessment is rated “Critical” under the current structured materials.

Cascade path

Propagation grouped into four incident phases (ordered along the cascade).

1 Pressure build-up

- **Hotspot traffic / localized pressure → shared resource** — Hotspot traffic and resource contention combined, driving queueing and connection waits upward in shared resource and forming the initial pressure wave.

2 Propagation diffusion

- **shared resource → middle-tier service** — When shared resource came under pressure, middle-tier service failed to form an effective shock-absorption layer through circuit breaking, backoff, or isolation, and buffer budget was rapidly consumed.

3 Cascading instability

- **middle-tier service → upstream entry layer** — Retries, fallback traffic, or synchronous coupling inside middle-tier service pushed the abnormal condition back into upstream entry layer, and the incident entered a clear amplification stage.

4 Recovery adjustment

- **upstream entry layer → core service path** — Because selective sacrifice actions were triggered too late, the upstream entry layer eventually showed elevated RT and error rates, and the core service path was materially compromised.

This version primarily relies on explicit dependencies, topology, or structured downstream-effect chains; it does not infer real call chains from raw metrics alone.

Dual-loop remediation

1 Inner loop — containment

- **Add circuit breaking, retry backoff, and concurrency limits at the stressed dependency boundary** — Case context: order-api. Cut the retry-driven amplification chain before local instability is pushed further upstream.
- **Introduce timeout budgets and fast-fail behavior for the stressed dependency** — Fail quickly when the dependency is under pressure so blocked workers and connections do not remain occupied.
- **Apply selective load-shedding and priority control at the entry layer** — Move selective sacrifice actions closer to the entry path so abnormal demand is contained earlier.

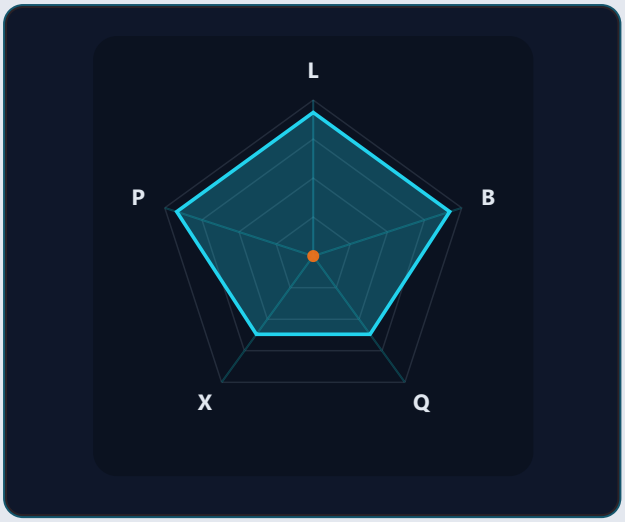
2 Outer loop — structural governance

- **Design isolation around contention-prone workloads and shared-resource pressure** — Use isolation, capacity headroom, and resource quotas around shared resources to weaken structural contention.
- **Separate shared-resource dependencies from the core service path** — Decouple the stressed dependency from the core path so a local fault cannot directly pierce the continuity boundary.
- **Institutionalize drills that combine peak load, shared resources, and retry policy** — Include peak-load pressure, shared-resource contention, and retry policy behavior in recurring drills to expose weak buffer and sacrifice points early.

3 **Resource focus**

Resource focus: prioritize investment in buffering, sacrifice, and isolation points rather than relying solely on post-incident capacity scaling.

Technical Appendix



Variable rules (summary)

Variable	Summary
L	Core Continuity — assess whether the core service path remains operable, recoverable, and maintainable. (Thresholds) core service s...
B	B is displayed with elevated-risk wording under evidence boundaries due to observed pressure persistence/headroom contraction.
Q	Queue and Accumulation Pressure — assess whether requests, tasks, connections, or resource usage show sustained buildup. (Thresholds)...
X	X is presented with an evidence-boundary posture and focuses on localized-yield timing/strength under observed signals.
P	Cascade Propagation Risk — assess whether a local issue can amplify along shared resources, dependency chains, or control chains. (Thresholds)...

Definitions support reading the five-variable chart; thresholds are indicative.

五变量详细判定

L 核心连续性 严重/临界

判定等级： 严重/临界

综合分值： 69

判定结论

核心连续性(L)判为严重:延迟/错误类共 2 项超过有效阈值,关键项包括 p99_latency_ms、error_ratio 等。

关键依据

- 关键指标与阈值对照:p99_latency_ms: 1380 vs 500 ms | threshold source: 输入提供 error_ratio: 0.0 vs 0.01 ratio | threshold source: 输入提供
- 命中规则:L-RT-001, L-RT-002, L-RT-ALR-001
- 阈值或比较基线:Core Continuity — assess whether the core service path remains operable, recoverable, and maintainable. (Thresholds) core service success rate >= 99.9% is healthy, 99% - 99.9% is warning, and < 99% is critical.
- 关注要点:Check core success rate, core-path RT, and user-visible impact first. If the main path is clearly breached, rate L as critical directly.

详细说明

证据要点:metrics, alerts - p99_latency_ms: 1380 (threshold 500) ms threshold source: 输入提供 - error_ratio: 0.0 (threshold 0.01) ratio threshold source: 输入提供 - p99_latency_ms: 1380 > 500 - error_ratio: 0.0 > 0.01

解释与关注:Check core success rate, core-path RT, and user-visible impact first. If the main path is clearly breached, rate L as critical directly.

B 缓冲承压容量

严重/临界

判定等级： 严重/临界

综合分值： 75

判定结论

Current evidence shows materially elevated pressure on buffer capacity, and the B display is kept at an elevated-risk posture under evidence boundaries.

关键依据

- Observed pressure persistence/headroom contraction supports an elevated B display posture.
- The wording is aligned with the finalized B grade and avoids diluted 'monitor-only' tone.

关注点 / 限制

Prioritize actions that directly restore effective headroom in the immediate response window. Track whether pressure de-escalates across consecutive windows after mitigation.

Q 排队与堆积压力

警告

判定等级： 警告

综合分值： 40.1

判定结论

压力堆积(Q)判为警告:队列/积压类 1 项超阈值,以 queue_depth 等为代表,堆积压力升高。

关键依据

- 关键指标与阈值对照:queue_depth: 1204 vs 800 count | threshold source: 输入提供
- 命中规则:Q-RT-001
- 阈值或比较基线:Queue and Accumulation Pressure — assess whether requests, tasks, connections, or resource usage show sustained buildup. (Thresholds) queue or accumulation utilization >= 85% is critical, 70% - 85% is high, and < 70%
- 关注要点:If queueing, backlog, or hotspot contention keeps rising and cannot self-recover, Q should be escalated.

详细说明

证据要点:metrics, alerts - queue_depth: 1204 (threshold 800) count threshold source: 输入提供 - queue_depth: 1204 > 800

解释与关注:If queueing, backlog, or hotspot contention keeps rising and cannot self-recover, Q should be escalated.

X 局部让渡强度

警告

判定等级： 警告

综合分值： 48

判定结论

Current materials indicate localized-yield signals that require continued attention to timing and strength; the statement remains evidence-bound.

综合判定说明

The overall rating remains “Warning”. Sentences that mention threshold pressure, saturation, or tight headroom are local observability excerpts; escalation requires the variable’s composite gating rules (coverage, persistence, and cross-signal aggregation), not a single metric line alone. The current grade means those escalation conditions are not collectively met under the present evidence boundary.

关键依据

- Track whether localized-yield actions are triggered early enough in the key response window.
- Track whether action intensity matches pressure dynamics in subsequent windows.

关注点 / 限制

Current materials indicate localized-yield signals that require continued attention to timing and strength; the statement remains evidence-bound.

P 级联扩散潜势

严重/临界

判定等级：严重/临界

综合分值：85

判定结论

Focus service context: “order-api”. The cascade-propagation assessment is rated “Critical” under the current structured materials. Where topology or downstream-impact objects are incomplete, the narrative stays conservative and does not extrapolate a full multi-hop chain beyond the evidence boundary.

关键依据

- 关键指标与阈值对照:Boundary: this version primarily relies on explicit dependencies, topology, or structured downstream-effect chains; it does not infer real call chains from raw metrics alone. order-api order-api(无下游 hop;若仅单 scope 或无超阈对齐
- 命中规则:P-RT-001, P-RT-ALR-001
- 阈值或比较基线:Cascade Propagation Risk — assess whether a local issue can amplify along shared resources, dependency chains, or control chains. (Thresholds) propagation potential >= 4x is critical, 2x - 4x is high, and < 2x is war
- 关注要点:When retries, fallback traffic, and synchronous coupling appear together, P should be treated as high risk or critical.

详细说明

级联扩散维度的结论以上述关键依据与综合等级为准;不作超出当前结构化材料的传播链扩张表述。